

Amin Adineh

Applied Physics
Computer Science & AI/ML

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PROFESSIONAL SUMMARY

I am Applied Physicist with expertise in Machine Learning, Artificial Intelligence, and Physics, specializing in Solid State Physics and Quantum Physics, Astrophysics and characterization methods in physics of X-ray and Batteries and microelectronics also in using AI/ML in Physics and integrating intelligent algorithms into software, hardware, robotic systems/simulations, and autonomous platforms. Experienced in developing scalable AI solutions that bridge scientific research with real-world engineering applications, optimization, and intelligent automation.

EDUCATION

Master of Science in Applied Physics

2025

Brandenburg University of Technology Cottbus-Senftenberg

Thesis: Temperature-Dependent Magnetotransport Characterization of Si/SiGe-Based Field Effect Stacks.

Research Interests: Quantum Transport, Semiconductor Physics, Machine Learning Integration, Scientific Automation.

Relevant Courses: Solid State Theory, Thermodynamics, Nanoelectronics, Renewable Energy, Semiconductor Technology, Characterization of Micro- and Nanomaterials, Light and Matter: Interaction in Nanostructures, Advanced Laboratory Techniques and Metrology, Optimization in Business Transformations.

Summer Courses: Attended two summer courses at BTU University related to Battery Technology and 2D Materials.

Independent Study: Basic knowledge of laboratory-based and synchrotron X-ray diffraction (XRD) techniques and material characterization methods for rechargeable batteries.

Advanced Coursework in Optimization and Operations Research

Weekly assignments on mathematical optimization methods: modeling with LP, IP, and MILP.

Exact algorithms: Branch-and-Bound, Dynamic Programming, decomposition techniques.

LP relaxation, primal-dual bounds, optimality gaps, combinatorial optimization.

Bachelor of Science in Applied Physics

2015 – 2020

Arak University

Relevant Courses: Lab-based X-rays and synchrotron radiation, XRD, Optics, Electromagnetism, Quantum Mechanics, Analytical Mechanics, Experimental Physics.

EXPERIENCE

DLR SchoolLab, BTU Cottbus-Senftenberg

Oct 2023 – 2026

DLR SchoolLab, BTU Cottbus, Germany

Supervisor: Prof. Dr. rer. pol. Lin Xie [\[Profile\]](#)

- **Assistant – Computer Science & AI/ML (Applied Machine Learning for Optimization (AI/ML))** Oct 2025 – Sep 2026
 - Developed a discrete-event RMFS simulation (MiniRawsim-inspired) for warehouse robotics, logistics workflows, and autonomous fleet behavior. [Project Link](#)
 - Applied Operations Research and optimization algorithms for task allocation, multi-robot scheduling, path planning, inventory management, and resource optimization.
 - Trained and deployed ML/CV models for robotic perception, object detection, scene understanding, decision-making, and autonomous interaction.
 - Integrated GPU-accelerated LLMs (TinyLlama, Qwen, GPT) into robotic systems for embodied AI, reasoning, and natural language interaction.
 - Developed optimization algorithms for storage, order/replenishment batching, pod repositioning, station activation, and robot coordination.

- Built analytics software and GUIs for robot dialogs, telemetry, behavioral analysis, and real-time system monitoring.
- Created agentic AI workflows using LangChain and LangGraph for multi-agent coordination, automation, and intelligent decision support.

Exhibition Presenter DLR (BTU) – AI Systems (Humanoid Robotics)

2025 – Present

- Presented humanoid robotics systems, including Ameca (AMI/AZI Desktop) and Azi (an altered version of the Ameca robot), at public technology events across Germany, demonstrating AI-driven robotics, human-robot interaction, and LLM integration at events such as Zukunftstag Brandenburg, Potsdamer Tag der Wissenschaften, Dresden MINT Festival, and IdeenExpo 2026, Der Tag der offenen Tür der Senftenberg BTU 2026. Supported live demonstrations, technical setup, and audience engagement with researchers, students, and industry professionals.

NASA Space Apps Challenge – AI Exoplanet Discovery

OCT 2025

- Developed an AI-assisted system for identifying and exploring exoplanet candidates using NASA and public astronomical datasets.
- Integrated an LLM to retrieve, analyze, and summarize scientific data, enabling natural language exploration and AI-driven candidate evaluation.
- Combined astronomical archives to support AI-based insights and exoplanet candidate analysis.
- Collaborated in a multidisciplinary team during the NASA Space Apps Challenge to design and prototype the solution.
- Project Link: <https://lnkd.in/eCFWU5GR> Team Profile: <https://lnkd.in/eD65vsVs>

IHP – Leibniz Institute for High Performance Microelectronics

Sep 2024 – Nov 2025

Frankfurt (Oder), Germany | Supervisor: Prof. Dr. rer. nat. habil. Christian Wenger [\[Profile\]](#)

• Master Thesis – Quantum Transport Research

Sep 2025 – Nov 2025

- Temperature-dependent magnetotransport characterization (1.5 mK to 10 mK).

• Research Project – AI & Cryogenic Data Analysis

Mar 2025 – Aug 2025

- Developed Python-based analysis pipeline for cryogenic measurements.
- Built real-time visualization for Hall and gate sweep experiments.
- Extracted Hall parameters (n , μ).
- Automated QHE data analysis and reporting.
- Applied AI embeddings for pattern recognition.

• Internship – Hardware-Software Integration

Sep 2024 – Mar 2025

- Integrated cryostat APIs using Python and NI LabVIEW (VI).
- Worked with Oxford Teslatron PT and L-Type Rapid systems.
- Conducted data acquisition in near-zero Kelvin environments.

SKILLS

AI Development:	LLMs, Generative AI, NLP Pipelines, RAG Systems, LangChain, LangGraph, Tokenization, Word Embeddings (Word2Vec, FastText, Transformer-based embeddings)
Robotics:	RMFS Simulation, Humanoid Systems, Human-Robot Interaction, Robot Behavior Analysis
Software Eng.:	MERN Stack, FastAPI, REST APIs, WebSockets, GUI Development, AI-based Software Dev.
Scientific Comp.:	Python, LabVIEW, Pandas, SciPy, Jupyter
Optimization:	LP, IP, MILP, Branch-and-Bound, Dynamic Programming, Heuristics, Metaheuristics, Approximation Algorithms, Network Optimization, Flow Algorithms
Core Competencies:	Scientific Automation, Data Analysis, Optimization Modeling, Operations Research
Characterization:	Basic knowledge of laboratory-based and synchrotron XRD techniques
Tools & Systems:	Git, Docker, Google Cloud, LaTeX, GPU Computing

LANGUAGES

English (C1 – Advanced) · German (B2 – Upper Intermediate) · Persian/Farsi (Native)